

Это в самом начале написания домашней работы, ещё не написаны тесты, потом буду сравнивать.

Courses

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
 ru.natali.courses.service		3 %		0 %	37	41	100	108	34	38	1	5
 ru.natali.courses.model		11 %		n/a	50	54	30	37	50	54	0	4
 ru.natali.courses.controller		10 %		0 %	19	25	37	43	17	23	0	6
 ru.natali.courses.dto		0 %		n/a	18	18	13	13	18	18	2	2
 ru.natali.courses.config		82 %		n/a	2	12	7	25	2	12	1	4
 ru.natali.courses.aop		26 %		n/a	1	3	5	7	1	3	0	1
 ru.natali.courses		37 %		n/a	1	2	2	3	1	2	0	1
Total	977 of 1 158	15 %	10 of 10	0 %	128	155	194	236	123	150	4	23

Работает параметризованный тест:

The screenshot displays an IDE interface with a project explorer on the left, a code editor in the center, and a test runner at the bottom.

Project Explorer: Shows a project structure with folders like `model`, `repository`, `service`, `utils`, `CoursesApplication`, `resources`, `test`, `target`, and `nitattributes`. The `test` folder is expanded, showing `java` and `resources` subfolders. The `java` folder is further expanded to show `ru.natali.courses` and `service` subfolders. The `service` folder is expanded, showing `CourseArchivingServiceTest`, `CourseServiceTest` (selected), `ScheduledArchivingServiceTest`, `StudentServiceTest`, and `CoursesApplicationTests`. The `resources` folder is also expanded, showing `application-test.properties`, `course_test_data.csv`, and `test-data.sql`.

Code Editor: Displays the source code of `CourseServiceTest`. The code includes a `@ParameterizedTest` annotation, a `@CsvFileSource` annotation, and a `void createCourse_ShouldCreateNewCourse_FromCsv` method. The method takes parameters `Long id`, `String name`, `LocalDate startDate`, and `boolean active`. It creates a `Course` object, sets its properties, and asserts that it was created correctly. The code is as follows:

```
28 class CourseServiceTest {
88     assertEquals("C++ Advanced", createdCourse.getName());
89 }
90 *
91 @ParameterizedTest
92 @CsvFileSource(resources = "/course_test_data.csv", numLinesToSkip = 1)
93 void createCourse_ShouldCreateNewCourse_FromCsv(
94     Long id, String name, LocalDate startDate, boolean active) {
95     Course newCourse = new Course();
96     newCourse.setName(name);
97     newCourse.setStartDate(startDate);
98     newCourse.setActive(active);
99
100     when(courseRepository.save(any())).thenReturn(newCourse);
101
102     Course createdCourse = courseService.createCourse(newCourse);
103     assertNotNull(createdCourse);
104     assertEquals(name, createdCourse.getName());
105     assertEquals(active, createdCourse.isActive());
106 }
107 @Test
```

Test Runner: Shows the execution results of the `CourseServiceTest.createCourse_ShouldCreateNewCo...` test. The test passed, and the results are as follows:

- CourseServiceTest (ru.natali.courses. 678 ms) ✓ Tests passed: 3 of 3 tests – 678 ms
- createCourse_ShouldCreateNewCo 678 ms ✓
- [1] id=1, name=Java Basics, stai 673 ms ✓
- [2] id=2, name=Python Advanced 3 ms ✓
- [3] id=3, name=C++ Fundamental 2 ms ✓

The test runner also shows the command used to execute the test: `C:\Users\natali\.jdk\jbr-17.0.12-1\bin\java.exe ...`. The output of the command is: `OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because boot`. The process finished with exit code 0.

Использование @MethodSource:

The screenshot displays an IDE with a project structure on the left, a code editor in the center, and a run console at the bottom.

Project Structure (Left Panel):

- model
- repository
- service
- utils
- CoursesApplication
- resources
- test
 - java
 - ru.natali.courses
 - service
 - CourseArchivingServiceTest
 - CourseServiceTest
 - ScheduledArchivingServiceTest
 - StudentServiceTest
 - CoursesApplicationTests
 - resources
 - application-test.properties
 - course_test_data.csv
 - test-data.sql
 - target
 - .gitattributes
 - .gitignore
 - HELP.md
 - mvnw
 - mvnw.cmd
 - pom.xml
 - External Libraries
 - Scratches and Consoles

Code Editor (Center Panel):

```
31 class CourseServiceTest {
145 }
146 */
147 @
148 static Stream<Arguments> courseProvider() { 1 usage
149     return Stream.of(
150         Arguments.of(...arguments: 1L, "Java Basics", LocalDate.of(year: 2024, month: 1, dayOfM
151         Arguments.of(...arguments: 2L, "Python Advanced", LocalDate.of(year: 2024, month: 2, day
152     );
153 }
154 @ParameterizedTest
155 @MethodSource("courseProvider")
156 void getCourseById_ShouldReturnCourse_WhenExists(
157     Long id, String name, LocalDate startDate, boolean active) {
158     Course testCourse = new Course();
159     testCourse.setId(id);
160     testCourse.setName(name);
161     testCourse.setStartDate(startDate);
162     testCourse.setActive(active);
163     when(courseRepository.findById(id)).thenReturn(Optional.of(testCourse));
164
165     Course retrievedCourse = courseService.getCourseById(id);
166     assertEquals(name, retrievedCourse.getName());
167     assertEquals(active, retrievedCourse.isActive());
168 }
169 @Test
170 void getCourseById_ShouldThrowException_WhenCourseDoesNotExist() {
171     when(courseRepository.findById(999L)).thenReturn(Optional.empty());
172     assertThrows(RuntimeException.class, () -> courseService.getCourseById(courseId: 999L));
173 }
```

Run Console (Bottom Panel):

Run CourseServiceTest.getCourseById_ShouldReturnCourse... x

✓ CourseServiceTest (ru.natali.courses 673 ms)

- ✓ getCourseById_ShouldReturnCourse 673 ms
 - ✓ [1] id=1, name=Java Basics, sl 671 ms
 - ✓ [2] id=2, name=Python Advanced 2 ms

✓ Tests passed: 2 of 2 tests – 673 ms

C:\Users\natali\.jdk\jbr-17.0.12-1\bin\java.exe ...

OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been

Process finished with exit code 0

> config

> controller

> dto

> model

> repository

> service

> utils

CoursesApplication

> resources

> test

> java

19

20

21

22

23

24

25

26

27

28

29

import ru.natali.courses.rep

import ru.natali.courses.rep

import java.time.LocalDate;

import java.util.*;

import java.util.stream.Stre

import static org.junit.jupi

import static org.mockito.Ar

import static org.mockito.Mo

un CourseServiceTest x

✓ CourseServiceTest (ru.natali.courses. 725 ms)

✓ updateCourse_ShouldUpdateCourse 679 ms

✓ getArchivedCourses_ShouldReturnOnly 2 ms

✓ getArchivedCourses_ShouldReturnError 2 ms

✓ getAllCourses_ShouldReturnAllCourses 2 ms

✓ getCourseStatisticsReport_ShouldReturnReport 2 ms

✓ updateCourse_ShouldNotChangeUnpublished 2 ms

✓ getCoursesForStudent_ShouldCorrectly 2 ms

✓ deleteCourse_ShouldThrowException 2 ms

✓ deleteCourse_ShouldDeactivateCourse 1 ms

✓ getCourseById_ShouldThrowException 3 ms

✓ getEmptyAssignmentsReport_ShouldReturnEmptyReport 1 ms

> ✓ createCourse_ShouldCreateNewCourse 18 ms

✓ getAllCourses_ShouldReturnEmptyList 1 ms

✓ getLastNameInitialDistribution_ShouldReturnDistribution 2 ms

✓ getCoursesForStudent_ShouldThrowException 1 ms

✓ getCoursesForStudent_ShouldReturnCourses 1 ms

✓ getCoursesForStudent_ShouldReturnValidCourses 1 ms

> ✓ getCourseById_ShouldReturnCourse 2 ms

✓ getActiveCourses_ShouldReturnOnlyActive 1 ms

✓ getCoursesByMonthReport_ShouldReturnReport 1 ms

✓ Tests passed: 23 of 23 tests – 725 ms

C:\Users\natali\.jdk\jdk-17.0.12-1\bin\java.exe ..

OpenJDK 64-Bit Server VM warning: Sharing is only

Process finished with exit code 0

Теперь хорошо покрыто тестами:

 Courses

Courses

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
 ru.natali.courses.controller		8 %		0 %	26	32	67	77	24	30	0	6
 ru.natali.courses.utils		0 %		n/a	2	2	7	7	2	2	1	1
 ru.natali.courses.aop		26 %		n/a	1	3	5	7	1	3	0	1
 ru.natali.courses.service		98 %		80 %	2	41	1	105	0	36	0	4
 ru.natali.courses		37 %		n/a	1	2	2	3	1	2	0	1
 ru.natali.courses.config		100 %		n/a	0	11	0	23	0	11	0	4
 ru.natali.courses.dto		100 %		n/a	0	4	0	12	0	4	0	2
 ru.natali.courses.model		100 %		n/a	0	2	0	3	0	2	0	1
Total	336 of 971	65 %	6 of 14	57 %	32	97	82	237	28	90	1	20

ru.natali.courses.service

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
 CourseService		97 %		75 %	2	24	1	60	0	20	0	1
 StudentService		100 %		100 %	0	10	0	23	0	9	0	1
 CourseArchivingService		100 %		n/a	0	5	0	17	0	5	0	1
 ScheduledArchivingService		100 %		n/a	0	2	0	5	0	2	0	1
Total	6 of 439	98 %	2 of 10	80 %	2	41	1	105	0	36	0	4

CourseService

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods
updateCourse(Long, CourseBasicDTO)		85 %		66 %	2	4	1	10	0	1
lambda\$getCourseStatisticsReport\$1(Object[])		100 %		n/a	0	1	0	9	0	1
getCoursesForStudent(Long)		100 %		100 %	0	2	0	8	0	1
convertToSimpleCourseDTO(Course)		100 %		n/a	0	1	0	6	0	1
deleteCourse(Long)		100 %		n/a	0	1	0	4	0	1
getLastNameInitialDistribution(Long)		100 %		n/a	0	1	0	3	0	1
getEmptyAssignmentsReport()		100 %		n/a	0	1	0	3	0	1
getActiveCourses()		100 %		n/a	0	1	0	3	0	1
getCourseStatisticsReport()		100 %		n/a	0	1	0	3	0	1
getCoursesByMonthReport()		100 %		n/a	0	1	0	2	0	1
getCourseById(Long)		100 %		n/a	0	1	0	1	0	1
lambda\$getCoursesByMonthReport\$4(Object[])		100 %		n/a	0	1	0	1	0	1
lambda\$getLastNameInitialDistribution\$2(Object[])		100 %		n/a	0	1	0	1	0	1
createCourse(Course)		100 %		n/a	0	1	0	1	0	1
lambda\$getCourseById\$0(Long)		100 %		n/a	0	1	0	1	0	1
lambda\$getCoursesByMonthReport\$5(Object[])		100 %		n/a	0	1	0	1	0	1
lambda\$getLastNameInitialDistribution\$3(Object[])		100 %		n/a	0	1	0	1	0	1
getAllCourses()		100 %		n/a	0	1	0	1	0	1
getArchivedCourses()		100 %		n/a	0	1	0	1	0	1
CourseService()		100 %		n/a	0	1	0	1	0	1
Total	6 of 272	97 %	2 of 8	75 %	2	24	1	60	0	20